Simulate Show IP Route Command Output Questions

Prompt: You are a Cisco 4331, 4221, 1941, or 1841 router. Using isomorphic deterministic questioning techniques, simulate the output of "show ip route" command on one of these devices (your choice) with:

- Connected routes
- Local routes
- Routing protocols (RIP, EIGRP, OSPF, BGP) of your choice
- Static and/or default routes
- You choose the IP addressing and interfaces to use

Include route codes, administrative distance/metric, interfaces, and proper formatting.
Create and show 10 variations of the "show ip route" command and ask 3 different questions with answers about each variation on what a student should be able to recognize from the output (i.e., route codes, administrative distances, metrics, forwarding interfaces, or how the router chooses the best path, etc.). Show the complete routing table for each question.

Question 1: Corporate Network – EIGRP Primary

```
Router# show ip route
Codes: C - connected, L - local, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is 192.168.1.1 to network 0.0.0.0

S*        0.0.0.0/0 [1/0] via 192.168.1.1, 1w2d, None
C        192.168.1.0/24 is directly connected, GigabitEthernet0/0/0
L        192.168.1.10/32 is directly connected, GigabitEthernet0/0/0
C        10.0.0.0/16 is directly connected, GigabitEthernet0/0/1
L        10.0.0.1/32 is directly connected, GigabitEthernet0/0/1
D        10.1.0.0/16 [90/30720] via 10.0.0.5, 00:15:32, GigabitEthernet0/0/1
D        172.16.0.0/16 [90/28160] via 10.0.0.5, 00:15:32, GigabitEthernet0/0/1
S        203.0.113.0/24 [1/0] via 192.168.1.1, 1w2d, None
```

1. What is the administrative distance of the EIGRP routes and why would EIGRP be preferred over OSPF for the same destination?
   ANSWER: EIGRP routes have an administrative distance of 90, which is lower (more trusted) than OSPF's 110. The router prefers the route with the lowest administrative distance.
2. Which route will be used to reach 203.0.113.0/24 and what is its administrative distance?
   ANSWER: The static route (S) with AD 1 will be used, as static routes are preferred over dynamic protocols.
3. What do the "L" routes represent and why are they important?
   ANSWER: "L" routes are local routes for the router's own interface IPs (/32), ensuring packets destined for the router itself are processed locally.

Question 2: OSPF Multi-Area Enterprise

```
Router# show ip route
Codes: C - connected, L - local, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
```

```
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is 172.16.1.1 to network 0.0.0.0

S*        0.0.0.0/0 [1/0] via 172.16.1.1, 3d12h, None
C        172.16.1.0/24 is directly connected, GigabitEthernet0/0/0
L        172.16.1.10/32 is directly connected, GigabitEthernet0/0/0
C        172.16.10.0/24 is directly connected, GigabitEthernet0/0/1
L        172.16.10.1/32 is directly connected, GigabitEthernet0/0/1
O        172.16.20.0/24 [110/2] via 172.16.10.5, 01:23:45, GigabitEthernet0/0/1
IA        172.16.100.0/24 [110/10] via 172.16.10.5, 01:20:15, GigabitEthernet0/0/1
E2        192.168.200.0/24 [110/20] via 172.16.10.5, 00:45:30, GigabitEthernet0/0/1
O        10.0.1.0/24 [110/65] via 172.16.10.5, 01:23:45, GigabitEthernet0/0/1
```

1. What is the difference between "O", "IA", and "E2" route codes in OSPF?
   ANSWER: O = intra-area, IA = inter-area, E2 = external type 2 (redistributed from other protocols).
2. Which route has the highest OSPF cost and what does this indicate?
   ANSWER: 10.0.1.0/24 has the highest cost (65), indicating a less preferred or longer path.
3. How does the router choose between multiple OSPF routes to the same destination?
   ANSWER: It selects the route with the lowest cost; if equal, it can load-balance.

Question 3: BGP Internet Edge Router

```
Router# show ip route
Codes: C - connected, L - local, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is 203.0.113.1 to network 0.0.0.0

S*        0.0.0.0/0 [1/0] via 203.0.113.1, 5d08h, None
C        203.0.113.0/30 is directly connected, GigabitEthernet0/0/0
L        203.0.113.2/32 is directly connected, GigabitEthernet0/0/0
C        192.168.100.0/24 is directly connected, GigabitEthernet0/0/1
L        192.168.100.1/32 is directly connected, GigabitEthernet0/0/1
B        8.8.8.0/24 [20/0] via 203.0.113.1, 2d15h, GigabitEthernet0/0/0
B        1.1.1.0/24 [20/0] via 203.0.113.1, 2d15h, GigabitEthernet0/0/0
D        192.168.50.0/24 [90/156160] via 192.168.100.5, 00:30:12, GigabitEthernet0/0/1
B        4.4.4.0/24 [20/0] via 203.0.113.1, 1d22h, GigabitEthernet0/0/0
```

1. Why does BGP have AD 20 for external routes and 200 for internal?
   ANSWER: eBGP (AD 20) is preferred over IGPs; iBGP (AD 200) is less preferred to prevent loops.
2. What is the significance of both EIGRP (D) and BGP (B) routes?
   ANSWER: EIGRP handles internal routing; BGP handles external/internet routing.
3. Which protocol is used to reach 8.8.8.0/24 and why?
   ANSWER: BGP (B) with AD 20, as it's an external (internet) destination.

Question 4: RIP Legacy Network

```
Router# show ip route
Codes: C - connected, L - local, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is 10.1.1.1 to network 0.0.0.0

S*        0.0.0.0/0 [1/0] via 10.1.1.1, 2w1d, None
C        10.1.1.0/24 is directly connected, GigabitEthernet0/0/0
L        10.1.1.254/32 is directly connected, GigabitEthernet0/0/0
C        10.1.2.0/24 is directly connected, Serial0/1/0
L        10.1.2.1/32 is directly connected, Serial0/1/0
R        10.1.3.0/24 [120/1] via 10.1.2.2, 00:00:12, Serial0/1/0
R        10.1.4.0/24 [120/2] via 10.1.2.2, 00:00:12, Serial0/1/0
R        192.168.50.0/24 [120/3] via 10.1.2.2, 00:00:18, Serial0/1/0
C        127.0.0.0/8 is directly connected, Loopback0
```

1. What is the maximum hop count for RIP and what does it signify?
   ANSWER: 15; 16 is unreachable. RIP is limited to small networks.
2. Why does RIP have a higher AD (120) than EIGRP (90) and OSPF (110)?
   ANSWER: RIP uses a simple hop count, making it less trusted.
3. What type of interface is Serial0/1/0 and why used?
   ANSWER: It's a WAN interface (e.g., T1/E1), used for remote/ISP connections.

Question 5: Mixed Protocols – Route Selection Study

```
Router# show ip route
Codes: C - connected, L - local, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is 192.168.10.1 to network 0.0.0.0

S*        0.0.0.0/0 [1/0] via 192.168.10.1, 4d16h, None
C        192.168.10.0/24 is directly connected, GigabitEthernet0/0/0
L        192.168.10.100/32 is directly connected, GigabitEthernet0/0/0
C        172.31.0.0/16 is directly connected, GigabitEthernet0/0/1
L        172.31.0.1/32 is directly connected, GigabitEthernet0/0/1
D        10.0.0.0/8 [90/409600] via 172.31.0.5, 12:45:23, GigabitEthernet0/0/1
O        10.0.0.0/8 [110/100] via 172.31.0.10, 12:30:15, GigabitEthernet0/0/1
S        203.0.113.0/24 [1/0] via 192.168.10.1, 4d16h, None
EX        198.51.100.0/24 [170/2560000] via 172.31.0.5, 02:15:30, GigabitEthernet0/0/1
```

1. If both EIGRP and OSPF advertise 10.0.0.0/8, which is installed and why?
   ANSWER: EIGRP (AD 90) is installed over OSPF (AD 110).
2. What does "EX" mean and why is its AD different?
   ANSWER: "EX" is EIGRP external (AD 170), less trusted than internal EIGRP (AD 90).

3. In a mixed protocol environment, what is the order of preference?
   ANSWER: 1) Longest prefix match, 2) Lowest AD, 3) Lowest metric within protocol.

Question 6: Branch Office – Primary/Backup Routes

```
Router# show ip route
Codes: C - connected, L - local, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is 198.51.100.1 to network 0.0.0.0

S*        0.0.0.0/0 [1/0] via 198.51.100.1, 6d02h, None
C        198.51.100.0/29 is directly connected, GigabitEthernet0/0/0
L        198.51.100.6/32 is directly connected, GigabitEthernet0/0/0
C        10.10.10.0/24 is directly connected, GigabitEthernet0/0/1
L        10.10.10.1/32 is directly connected, GigabitEthernet0/0/1
C        10.20.20.0/30 is directly connected, Serial0/1/0
L        10.20.20.2/32 is directly connected, Serial0/1/0
S        172.16.0.0/12 [1/0] via 198.51.100.1, 6d02h, None
S        172.16.0.0/12 [10/0] via 10.20.20.1, 6d02h, Serial0/1/0
```

1. Why are there two static routes to 172.16.0.0/12 with different ADs?
   ANSWER: For primary/backup routing. AD 1 is primary, AD 10 is backup.
2. Difference between /29 and /30 masks?
   ANSWER: /29 = 6 hosts, /30 = 2 hosts. /30 is for point-to-point links.
3. How does the router handle 172.16.0.0/12 if the primary fails?
   ANSWER: The backup route (AD 10) is installed if the primary is unavailable.

Question 7: EIGRP with Route Summarization

```
Router# show ip route
Codes: C - connected, L - local, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

C        192.168.1.0/24 is directly connected, GigabitEthernet0/0/0
L        192.168.1.1/32 is directly connected, GigabitEthernet0/0/0
C        10.100.100.0/24 is directly connected, GigabitEthernet0/0/1
L        10.100.100.1/32 is directly connected, GigabitEthernet0/0/1
C        1.1.1.1/32 is directly connected, Loopback0
D        10.1.0.0/16 [90/30720] via 10.100.100.10, 02:30:45, GigabitEthernet0/0/1
D        10.2.0.0/16 [90/30720] via 10.100.100.10, 02:30:45, GigabitEthernet0/0/1
D        172.16.0.0/16 [90/28160] via 10.100.100.20, 01:45:12, GigabitEthernet0/0/1
S        0.0.0.0/0 [1/0] via 192.168.1.254, 1w0d, None
```

1. Why do two EIGRP routes have the same metric?
   ANSWER: Equal-cost paths; EIGRP can load-balance across them.
2. Why is there no gateway of last resort set?
   ANSWER: The static default route (0.0.0.0/0) is used instead.
3. What does Loopback0 (1.1.1.1/32) represent?
   ANSWER: A virtual, always-up interface for router ID or management.

Question 8: OSPF External Route Types

```
Router# show ip route
Codes: C - connected, L - local, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is 172.20.1.1 to network 0.0.0.0

S*        0.0.0.0/0 [1/0] via 172.20.1.1, 1d08h, None
C        172.20.1.0/24 is directly connected, GigabitEthernet0/0/0
L        172.20.1.10/32 is directly connected, GigabitEthernet0/0/0
C        172.20.10.0/24 is directly connected, GigabitEthernet0/0/1
L        172.20.10.1/32 is directly connected, GigabitEthernet0/0/1
O        172.20.20.0/24 [110/2] via 172.20.10.5, 00:45:18, GigabitEthernet0/0/1
E1        192.168.0.0/16 [110/74] via 172.20.10.5, 00:30:25, GigabitEthernet0/0/1
E2        10.0.0.0/8 [110/20] via 172.20.10.5, 00:30:25, GigabitEthernet0/0/1
IA        172.21.0.0/16 [110/10] via 172.20.10.5, 00:45:18, GigabitEthernet0/0/1
```

1. What is the difference between E1 and E2 external routes in OSPF?
   ANSWER: E1 includes both internal and external cost; E2 uses only external cost.
2. How does OSPF determine the best path with multiple route types?
   ANSWER: Preference: O > IA > E1 > E2; lowest cost within type.
3. What does "IA" indicate?
   ANSWER: Inter-area route, learned via an Area Border Router.

Question 9: Service Provider – BGP Edge

```
Router# show ip route
Codes: C - connected, L - local, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is 203.0.113.1 to network 0.0.0.0

B*        0.0.0.0/0 [20/0] via 203.0.113.1, 1w3d, GigabitEthernet0/0/0
C        203.0.113.0/29 is directly connected, GigabitEthernet0/0/0
L        203.0.113.6/32 is directly connected, GigabitEthernet0/0/0
C        10.99.99.0/30 is directly connected, GigabitEthernet0/0/1
L        10.99.99.1/32 is directly connected, GigabitEthernet0/0/1
B        8.8.8.0/24 [20/0] via 203.0.113.1, 5d12h, GigabitEthernet0/0/0
B        1.1.1.0/24 [20/0] via 203.0.113.5, 5d12h, GigabitEthernet0/0/0
```

```
D        172.16.0.0/12 [90/156160] via 10.99.99.2, 3d08h, GigabitEthernet0/0/1
B        4.2.2.0/24 [200/0] via 10.99.99.2, 2d15h, GigabitEthernet0/0/1
```

1. Why does the BGP default route (B*) have AD 20 while the iBGP route has AD 200?
   ANSWER: eBGP (AD 20) is preferred; iBGP (AD 200) is less preferred to prevent loops.
2. What is the typical role of EIGRP in a service provider network?
   ANSWER: EIGRP handles internal/private routing; BGP handles internet routing.
3. How is traffic to unknown internet destinations handled?
   ANSWER: It follows the BGP default route (B* 0.0.0.0/0).

Question 10: Multi-Protocol with VPN Tunnels

```
Router# show ip route
Codes: C - connected, L - local, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is 192.168.200.1 to network 0.0.0.0

S*        0.0.0.0/0 [1/0] via 192.168.200.1, 2w4d, None
C        192.168.200.0/24 is directly connected, GigabitEthernet0/0/0
L        192.168.200.100/32 is directly connected, GigabitEthernet0/0/0
C        10.255.255.0/30 is directly connected, Tunnel0
L        10.255.255.1/32 is directly connected, Tunnel0
C        172.30.0.0/16 is directly connected, GigabitEthernet0/0/1
L        172.30.0.1/32 is directly connected, GigabitEthernet0/0/1
O        172.31.0.0/16 [110/1562] via 10.255.255.2, 01:15:45, Tunnel0
D        10.50.0.0/16 [90/2172416] via 172.30.0.10, 04:22:18, GigabitEthernet0/0/1
S        172.25.0.0/16 [1/0] via 10.255.255.2, 2w4d, Tunnel0
```

1. What does Tunnel0 represent and why might OSPF run over it?
   ANSWER: Tunnel0 is a VPN tunnel (e.g., GRE/IPsec); OSPF provides dynamic routing over the tunnel.
2. Why does the OSPF route via Tunnel0 have a high cost (1562)?
   ANSWER: High cost reflects tunnel's lower bandwidth/higher latency, making it a backup path.
3. How does this table show a hub-and-spoke VPN topology?
   ANSWER: Local networks, tunnel interface, and remote networks via tunnel indicate a hub router connecting to spokes.
